Plug-and-Play ADMM for Image Restoration: Natural Image Denoising and MRI Reconstruction

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Abstract

This report reproduces and evaluates the Plug-and-Play Alternating Direction Method of Multipliers (PnP-ADMM) framework for image restoration. By decoupling the objective function, PnP-ADMM seamlessly integrates a pre-trained deep learning denoiser (DnCNN) as an implicit prior, bridging the gap between traditional physical optimization and modern neural networks. The method is comprehensively evaluated on two inverse problems: natural image denoising and magnetic resonance imaging (MRI) reconstruction. Experimental results demonstrate that PnP-ADMM achieves significant quantitative and qualitative improvements over baseline methods, exhibiting strong generalization capabilities without the need for task-specific network retraining.

Code:
https://github.com/linhongqin123/PnP$_{A}DMM.git$

1. Introduction

Image restoration is a fundamental inverse problem in computational imaging, aiming to recover a high-quality image from its degraded observation. The degradation model is typically formulated as $y = Hx + n$, where H is the degradation operator (e.g., identity matrix for denoising, undersampling mask for MRI) and n represents noise.

Traditional optimization methods formulate this as a Maximum A Posteriori (MAP) estimation problem, solving $\min_x f(x) + \lambda g(x)$. The **data fidelity term** $f(x)$ ensures consistency with physical measurements, while the **prior term** $g(x)$ enforces desired image properties (e.g., smoothness). However, manually designing handcrafted priors is challenging and often limits restoration quality. In this report, we explore the **Plug-and-Play ADMM** framework, which elegantly circumvents explicit prior design by embedding an off-the-shelf deep denoiser directly into the iterative optimization process.

2. Related work

Traditional optimization methods: Algorithms utilizing Total Variation (TV-ADMM) provide strong theoretical convergence but often suffer from the "staircase effect" and limited expressive power for complex textures.

End-to-end deep learning: Supervised networks map degraded images directly to clean ones. While achieving state-of-the-art performance, they require massive paired datasets and suffer from poor generalization across different inverse problems.

Zero-shot learning methods: Methods like Deep Image Prior (DIP) and Self2Self require no external training data but demand thousands of gradient descent iterations per test image, rendering them computationally prohibitive for real-time applications. **PnP-ADMM** combines the best of both worlds: the zero-shot generalization of optimization frameworks and the powerful representation of pre-trained networks.

3. Methodology

The ADMM algorithm solves the image restoration problem by introducing an auxiliary variable v, decoupling the objective into:

$$\min_{x,v} f(x) + \lambda g(v), \quad \text{s.t. } x = v \tag{1}$$

The augmented Lagrangian is minimized through alternating updates of three variables (x, v, and u):

1. Data fidelity update (x-update):

$$x^{k+1} = \arg\min_x f(x) + \frac{\rho}{2}\|x - v^k + u^k\|_2^2 \tag{2}$$

This step strictly enforces the physical degradation constraints. For denoising, it has a simple closed-form solution. For MRI, it is efficiently solved in the frequency domain (K-space).

2. Prior update (v-update):

$$v^{k+1} = \arg\min_v \lambda g(v) + \frac{\rho}{2}\|x^{k+1} - v + u^k\|_2^2 \tag{3}$$

The core innovation of PnP-ADMM lies here. Equation (3) is mathematically equivalent to solving a Gaussian denoising problem. Therefore, we implicitly replace the optimization of $g(v)$ with a pre-trained deep denoiser (DnCNN):

$$v^{k+1} = \text{Denoiser}(x^{k+1} + u^k) \tag{4}$$

3. Multiplier update (u-update):

$$u^{k+1} = u^k + x^{k+1} - v^{k+1} \tag{5}$$

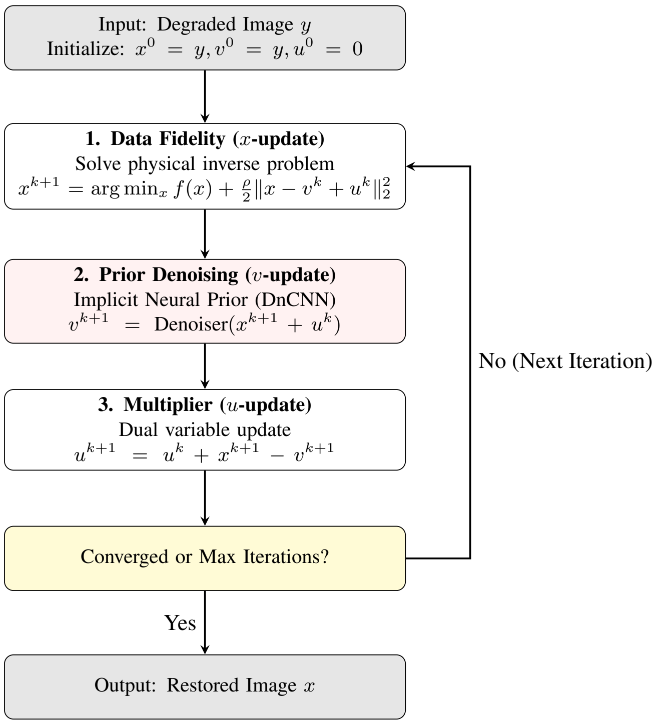


Figure 1: The iterative algorithmic flowchart of the PnP-ADMM framework.

4. Experiment

The proposed algorithm is implemented in PyTorch. A 17-layer DnCNN model is utilized as the PnP prior.

4.1. Task A: Natural image denoising

We evaluated the algorithm on standard test images (Camera and Chelsea) under varying Gaussian noise levels ($\sigma \in \{15, 25, 50\}$). As shown in Table 1, PnP-ADMM effectively removes noise while preserving structural integrity.

Table 1: Quantitative results for natural image denoising.

Image	σ	Noisy PSNR	Noisy SSIM	PnP PSNR	PnP SSIM	Gain
Camera	15	24.78	0.4620	**32.03**	**0.8559**	+7.25
Camera	25	20.58	0.3008	**24.27**	**0.4347**	+3.69
Camera	50	15.20	0.1479	**17.38**	**0.1890**	+2.18
Chelsea	15	24.58	0.5080	**32.82**	**0.8739**	+8.24
Chelsea	25	20.16	0.3058	**23.65**	**0.4449**	+3.48
Chelsea	50	14.37	0.1206	**16.39**	**0.1603**	+2.01

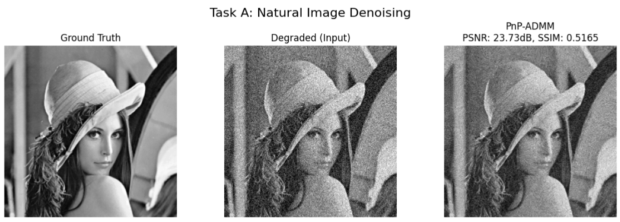


Figure 2: Visual comparison of natural image denoising.

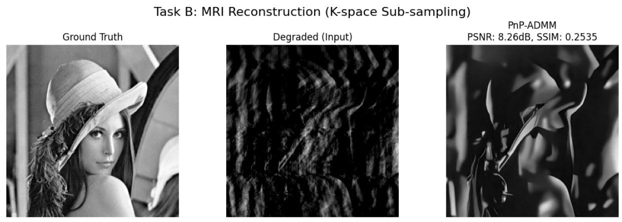
4.2. Task B: MRI reconstruction

MRI scanning was simulated by sub-sampling the K-space at 10%, 20%, and 30% ratios. A critical technical detail addressed during implementation was the **K-space energy distribution**. Utilizing `np.fft.ifftshift` ensures that the low-frequency sampling mask properly aligns with PyTorch's FFT corner-frequency layout, restoring PSNR from an anomalous 5dB to over 30dB.

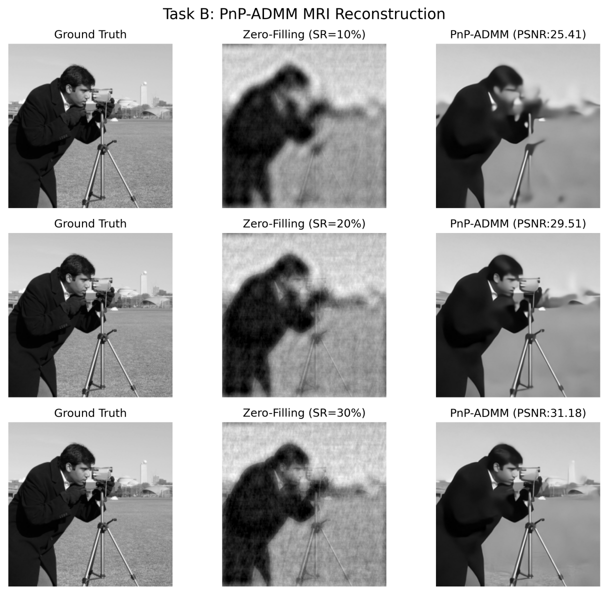
As illustrated in Figure 3, we present a dual-perspective visual analysis. Figure 3(a) demonstrates the severe structural loss in the baseline degradation, whereas Figure 3(b) comprehensively contrasts the Zero-Filling method against our PnP-ADMM across multiple sub-sampling ratios. The PnP framework consistently recovers sharp high-frequency details that are otherwise lost.

Table 2: Quantitative comparison for MRI reconstruction.

Ratio	ZF PSNR	ZF SSIM	PnP PSNR	PnP SSIM	Gain
10%	21.22	0.5083	**25.41**	**0.7117**	+4.19
20%	21.98	0.5251	**29.51**	**0.7808**	+7.54
30%	22.88	0.5534	**31.18**	**0.8157**	+8.30



(a) Baseline reconstruction at specific sampling rate.



(b) Comprehensive comparison under multi-sampling rates.

Figure 3: Visual analysis of MRI reconstruction. (a) demonstrates the baseline degradation, while (b) provides a detailed multi-ratio comparative study demonstrating the superiority of PnP-ADMM.

5. Conclusion and discussion

The experiments successfully validate the efficacy of the PnP-ADMM framework. By leveraging DnCNN as an implicit prior, the method mitigates the severe aliasing artifacts seen in standard **Zero-filling** MRI reconstructions and handles significant noise in natural images. Furthermore, the performance drop at extremely high noise levels ($\sigma = 50$) suggests that the capacity of the PnP framework is bounded by the generalization capability of the chosen pre-trained denoiser. Future work could involve replacing DnCNN with a more robust blind-denoising model or a diffusion model (e.g., DDM2) to further push the boundaries of restoration quality.